# The jet diffusion wake as moving geometry and three tests of its additional structure

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**Abstract**

CMS reports a charged-hadron depletion opposite the subleading jet in central lead–lead collisions, exceeding five standard deviations in its integrated low-momentum signal [1]. Fundamental Density Theory (FDT) accepts the corrected distributions and that significance bound on the source's own inputs and observable dictionary. The wake is the omnium's geometry resizing to the jet's energy flow; the curvature for the same stress tensor and data is general relativity's. The density disturbance and gravity's ripple are one stroke. The wake is visible only as the deviation of neighbouring patterns: a pattern crossing a density gradient at constant local speed is in free fall, its clocks and rulers rescaled together, and the source's gap and system subtractions make that comparison. Three prospective predictions test additional structure: an opposite-species toroidal-polarization intercept, a charge-odd annulus contrast, and a negative integrated contrast in proton–proton collisions. The first two change the observable dictionary through the magnetic projection of mechanical circulation; the third includes the mass shadow carried by a moving source. All are existence tests conditional by construction, with no signal magnitude assigned by the core. Covariance-aware decisions are evaluated under explicit illustrative noise models at zero signal and at nonzero test points. Four further candidates have no observable map, and the reported displacement of the depletion remains an in-sample reading.

## 1 Framework

The source status is experimental. Measured, observed and reported refer to source results; derived, estimated, reconstructed and illustrative-assumed identify the corresponding non-source statements. The analysis uses Core FDT Invariants, locked version 5.4, author lock 2026-09-13 [2], and the CMS result [1]. The identifiers J-nnn identify source claims at their physical L loci; U-nn identify non-source numerical inputs; NG-nn identify notation-guard rows, SA-nn the record windows where the source's data under-determine its reading, and DC-n the divergence candidates that each prediction or retirement answers. CMS records lead–lead (PbPb) and proton–proton (pp) collisions; these names label event selections throughout.

### 1.1 One medium, one chain, one stroke

Matter, energy and spacetime are one substance, the omnium (core 0.1). There are no separate particles and no boxes: there is only omnium and omnium patterns (0.1b). A hadron is a closed pattern on its own support. Around every point a sphere contains both higher- and lower-density regions. This nowhere-uniform density supplies interaction pairs everywhere. The clocks and rulers are themselves closed patterns; their adjustment is the geometry, with no background metric that they merely read (0.1a).

Let $E$ denote energy, $m$ a confined rest mass, $c$ the invariant local null-cone speed, $\varepsilon_0$ the electromagnetic permittivity and $\mu_0$ the permeability; $d$ and $t$ denote a stroke's length and duration. Core 0.1 fixes the quantisation chain

$$\frac{E}{m} = c^2 = \frac{1}{\varepsilon_0 \mu_0} = \left(\frac{d}{t}\right)^2 . \tag{1}$$

All terms adjust simultaneously. Quantised energy and fixed $c$ place massive discreteness in confinement eigenvalues and null discreteness in locked periods and paths. No variable in the chain vanishes. The constant $\hbar$ is reduced Planck action and $h$ is Planck action; $\omega$ denotes angular frequency, and $T_{\text{mode}}$ and $\ell_{\text{mode}}$ denote a mode's kept period and thread length. The circle constant is $\pi$. For a massive mode, core 0.1a gives

$$\begin{aligned} E_{\text{mode}} &= m_{\text{mode}} c^2 = \hbar \omega_{\text{mode}}, \\ T_{\text{mode}} &= \frac{2\pi}{\omega_{\text{mode}}} = \frac{h}{E_{\text{mode}}}, \qquad \ell_{\text{mode}} = c T_{\text{mode}} = \frac{h}{m_{\text{mode}} c}. \end{aligned} \tag{2}$$

The thread length is not the support radius. A null stroke keeps the ratio of its length to its duration equal to $c$; it does not acquire a rest mass by that statement.

Let $G$ denote the gravitational constant, $F$ force, $F_{\max}$ its ceiling, $\kappa$ the Einstein coupling, and $\ell_{\text{P}}$ and $t_{\text{P}}$ the Planck length and time. The infimum stroke has length $d_{\min}$ and duration $t_{\min}$. Core 0.2 and 0.1a give

$$\begin{aligned} F_{\max} = \frac{c^4}{4G}, \qquad \kappa = \frac{8\pi G}{c^4}, \qquad \kappa F_{\max} = 2\pi, \\ F_{\max} d_{\min} t_{\min} = \hbar, \qquad d_{\min} = c t_{\min}, \\ d_{\min}^2 = \frac{\hbar c}{F_{\max}} = \frac{4\hbar G}{c^3} = 4\ell_{\text{P}}^2, \\ d_{\min} = 2\ell_{\text{P}}, \qquad t_{\min} = 2t_{\text{P}}. \end{aligned} \tag{3}$$

These are one-sided infima approached by the least closed pattern and occupied by none. They do not define a lattice or an occupied minimum cell.

The chain identifies the speed of mass–energy exchange, the electromagnetic propagation constant and the medium's stroke; 0.1b supplies one medium. Therefore the photon, graviton, gluon, phonon and magnon are the same spin-one carrier stroke read in different density and geometry regimes (I-2). A carrier is a disturbance of a medium at that medium's speed. The source's gluons are the stroke standing inside a closed pattern's own region. Matter configurations do not enter this carrier identity. There is no mediation and no carrier exchange (I-2). No sound speed of the source medium is stated here: the record gives none.

## 1.2 Notation guard

The table below fixes the complete source-symbol census (NG-01–NG-15). Source glyphs in its first column are names in text, not operative symbols. The azimuth is $\Phi$; $\eta$ is pseudorapidity; $p$ is momentum with upright T denoting its transverse component; $\Delta$ denotes a finite difference. The upright labels j1, j2 and ch denote leading jet, subleading jet and charged track. The correlations $\mathcal{R}_{\mathrm{sym}}$ and $\mathcal{R}_{\mathrm{asym}}$ are yields per dijet pair per radian per unit pseudorapidity for small and large dijet gaps. The Greek labels $\mu, \nu$ denote tensor components, not scalar quantities. The energy $E_{\mathrm{NN}}$ is the energy per nucleon pair; $E_{\mathrm{T}}$ is a transverse energy. Reserved symbols introduced below retain their single core meanings.

| Source token and locus | Manuscript notation and meaning |
|---|---|
| Azimuth phi, L43 | $\Phi$, $\Delta\Phi^{\mathrm{ch,j1}}$, $\Delta\Phi_{\mathrm{j1,j2}}$; $\phi$ is helical phase. |
| Distance R, L82 | Distance parameter 0.4 in words; $R$ is a geometrical radius. |
| Rsym, Rasym, L188 | $\mathcal{R}_{\mathrm{sym}}$, $\mathcal{R}_{\mathrm{asym}}$; no italic radius is a yield. |
| Radical over sNN, L20 | $E_{\mathrm{NN}} = 5.02\,\mathrm{TeV}$; $s$ denotes stroke length. |
| pT and its qualifiers, L94 | $p_{\mathrm{T}}$, $p_{\mathrm{T}}^{\mathrm{ch}}$, $p_{\mathrm{T}}^{\mathrm{j1}}$, $p_{\mathrm{T}}^{\mathrm{j2}}$; italic $T$ is temperature, $T_{\mu\nu}$ stress, $T_{\mathrm{mode}}$ a kept period. |
| ET and E, L128 | $E_{\mathrm{T}} = E/\cosh\eta$; both are energies. |
| Eta and its differences, L134 | $\eta^{\mathrm{j1}}$, $\eta^{\mathrm{j2}}$, $\Delta\eta^{\mathrm{ch,j1}}$, $\Delta\eta_{\mathrm{j1,j2}}$; all rapidity-like quantities are distinguished from the rapidity of a massive hyperon. |
| Chi-squared, L128 | Written in words; $\chi$ is orientation or a decorated response susceptibility. |
| Relative momentum error, L118 | Written in words; $\delta$ denotes the core compression contrast. |
| Transverse/beam coordinates, L124 | Written in words; $x$ is the odds-enhancing variable, $y_0$ the diagnostic floor, and $z_1, z_2$ the three-sphere coordinates. |
| Jet1, jet2, L61 | Upright j1 and j2 labels, never new variables. |
| Anti-kT, L106 | Anti-$\mathrm{k_T}$ in text; $k$ is a wavenumber. |
| Standard deviations, L23 | Words or sd; $\sigma$ is a branch tag, $\sigma_{\mathrm{SB}}$ the radiation flux constant. |
| Photon gamma, L325 | Photon in words; $\gamma$ has only that meaning. |
| Dimension D; A+A; q-hat; eta/s; tau and nu-tau; radicals over s and sNN; 2D, 3D; b-jet, L223 and reference titles | Dimensionality in words; reference-title tokens are not operative symbols; $D$ is the confinement operator, $A$ mode amplitude, $q$ density order, $\tau$ stroke duration, $\nu$ decorated topological integers, and $b$ the core force fraction or its bounded-output application. |

The table's unused core labels are reservations, not additional observables. No torus-knot integers or reference-title transport coefficients are used. The symbol $\Omega$ below is exclusively a confinement domain; angular line elements use $\mathfrak{o}$. Optical indices use $\mathfrak{n}$; $n$ is exclusively helix count. Statistical thresholds use a multiplier $\mathfrak{z}$ and standard deviations, never $\sigma$ or the coordinate $z$.

## 1.3 Atlas, charts and optical containment

Let $\mathcal{M}$ denote total gravitating energy in mass units, $R_s$ its gravitational radius, $r_f$ a declared field-evaluation radius, $R_b$ a support radius and $r_c$ the interior cutoff. The branch tag $\sigma$ is positive on the exterior and negative on the interior. The dimensionless curvature ratio is $\alpha_\sigma$;

the positive odds are $X_\sigma$. The symbol $\infty$ denotes an unbounded exterior coordinate range. The odds $X_{\text{ext}}, X_{\text{int}}$ and ratios $\alpha_{\text{ext}}, \alpha_{\text{int}}$ use explicit exterior and interior labels. Core 0.3, 0.4 and I-3 give

$$\begin{aligned} R_s &= \frac{2G\mathcal{M}}{c^2}, \qquad \alpha_\sigma = \left(\frac{R_s}{r_f}\right)^\sigma, \\ \sigma &= +1: \ R_s < r_f < \infty, \qquad \sigma = -1: \ r_c \le r_f < R_s, \\ X_\sigma &= \frac{\alpha_\sigma}{1-\alpha_\sigma}, \qquad \alpha_\sigma = \frac{X_\sigma}{1+X_\sigma}, \qquad 0 < \alpha_\sigma < 1, \\ X_{\text{ext}} &= \frac{R_s}{r_f - R_s}, \qquad X_{\text{int}} = \frac{r_f}{R_s - r_f}, \qquad \frac{\partial \alpha_\sigma}{\partial r_f} = -\frac{\sigma \alpha_\sigma}{r_f}. \end{aligned} \tag{4}$$

These are ratios of two lengths, not fitted density scores. The limiting skin is approached as $\alpha_\sigma \to 1^-$ and the remote exterior or dilute interior end as $\alpha_\sigma \to 0^+$. The open range alone warrants no application.

The three-sphere is the support's shape, not a container for the omnium. Its two solid tori distinguish the phases of one substance. On A-1, the $\alpha_\sigma > 1/2$ solid torus is matter and closed braids; the $\alpha_\sigma < 1/2$ solid torus is decompressed spacetime. The Clifford torus is the A-1 seam at $\alpha_\sigma = 1/2$: exterior $r_f = 2R_s$ or interior $r_f = R_s/2$. The skin fold and the seam are distinct. Two truncated cones meet circle-to-circle at the limiting skin; the endpoints are Hopf circles, never points. Matter and spacetime are phases of the same omnium; the repeated form terminates in the unoccupied infimum of Eq. (3). The collision's exterior compactness at a laboratory radius reads its far field; it does not locate its closed hadron patterns on the matter torus.

For the chart inventory let $\beta_\sigma$ denote the computational squash, $q_\sigma$ the density-order proxy, $\alpha_{\text{ovl}}$ the overlap label, $\alpha_{\text{conf}}$ the confinement label and $\alpha_{\text{mode}}$ the mode label. Their corresponding odds, where named, use $X$ with the same label. The nuclear diagnostic $\alpha_{\text{core}}$ denotes only its interior skin saturation.

| Application | Chart inventory | Warrant and scope |
|---|---|---|
| A-1 | $(X_\sigma, \alpha_\sigma)$, Eq. (4) | Geometric; both branches tagged. |
| A-2 | $(\alpha_\sigma, \beta_\sigma)$; $\beta_\sigma = \alpha_\sigma/(1+\alpha_\sigma)$ | Geometric computational squash; no force, density or optical input. |
| A-3 | $(X_{\text{ovl}}, \alpha_{\text{ovl}})$ | PROVISIONAL token and open range only; no state value, side or crossing. |
| A-4 | $(X_{\text{conf}}, \alpha_{\text{conf}})$ | PROVISIONAL token and open range only; no confinement address. |
| A-5 | $\alpha_{\text{mode}}$ | PROVISIONAL naming only; no ordering or trajectory. |
| Skin | $\alpha_{\text{core}} \to 1^-$ | Interior saturation diagnostic, distinct from exterior compactness. |
| Order | $q_\sigma = \alpha_\sigma^{3\sigma}$ | Proxy, not a measured local density. |

The A-3 geometric boundary of I-1a is two thirds, distinct from the A-1 seam. No source medium is assigned to either side of that boundary. Thermal disorder remains on the attractive branch; a crossing needs density, not temperature. The words hot, dense, central and peripheral supply no compactness or overlap coordinate.

Let $\ell_{\mathrm{opt}}$ be optical path length and $d\mathfrak{o}^2$ the unit-sphere angular line element. The angles $\theta_1, \theta_2$ are incidence and refraction angles and $\mathfrak{n}_1, \mathfrak{n}_2$ the directional indices derived from the exact optical metric. The tensor $G_{\mu\nu}$ is Einstein curvature and $T_{\mu\nu}$ stress–energy; $\mu, \nu$ are spacetime component labels in this tensor context, not scalar quantities. Core I-3 has three faces, the atlas above and the two following equations:

$$d\ell_{\mathrm{opt}}^2 = \frac{dr_f^2}{(1-\alpha_{\mathrm{ext}})^2} + \frac{r_f^2 d\mathfrak{o}^2}{1-\alpha_{\mathrm{ext}}}, \qquad \mathfrak{n}_1 \sin\theta_1 = \mathfrak{n}_2 \sin\theta_2. \tag{5}$$

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \tag{6}$$

The optical metric is exact in its static areal chart; the Snell relation uses that metric's indices and is not a linearised arrow. The tensor equation is unchanged, with compactness downstream of its source and solution. Identical stress tensor, data and gauge give identical curvature. This is containment, including the wake's curvature.

Let $c_{\mathrm{eff}}$ be the effective chart aperture speed and $\mathfrak{n}_{\mathrm{cone}}$ its reciprocal speed ratio. The P-4 aperture candidate reads

$$c_{\mathrm{eff}} = c\sqrt{1-\alpha_\sigma}, \qquad \mathfrak{n}_{\mathrm{cone}} = (1-\alpha_\sigma)^{-1/2}. \tag{7}$$

It is not the invariant $c$ and is not identified with the sound speed of the record's medium. It does not replace Eq. (5).

### 1.4 The force identity and its domains

The inputs $\alpha_1, \alpha_2$ are the two geometrical curvature ratios of I-1. At every scale, its unconditional law and open-interval bound are

$$F = F_{\max}\alpha_1\alpha_2 = \frac{c^4}{4G}\alpha_1\alpha_2, \qquad 0 < \alpha_1, \alpha_2 < 1 \implies 0 < F < F_{\max}. \tag{8}$$

For the isolated, non-overlapping spherical pair, let $d$ be the centre separation, $\mathcal{M}_{\mathsf{i}}$ the gravitating masses and $\lambda_{\mathsf{i}} = \mathcal{M}_{\mathsf{i}}/d$ their mass-per-separation inputs, with $\mathsf{i}$ indexing the two sources in this paragraph. Cross-centre exterior evaluation sets $\alpha_{\mathsf{i}} = R_{s,\mathsf{i}}/d$. Core I-4's L1 identity is

$$F = \frac{G\mathcal{M}_1\mathcal{M}_2}{d^2} = G\lambda_1\lambda_2 = F_{\max}\alpha_1\alpha_2 = \frac{2\pi}{\kappa}\alpha_1\alpha_2. \tag{9}$$

The support conditions are $R_{b,\mathsf{i}} \geq R_{s,\mathsf{i}}$ and $d \geq R_{b,1} + R_{b,2}$. The arithmetic–geometric mean inequality then bounds the input product by one quarter, giving $F \leq F_{\max}/4$ on this pair domain alone, with equality at contact of saturated equal supports. This reduction does not restrict I-1. L2 is the weak-field, slow-motion reduction of general relativity; its leading Newtonian term has relative corrections of compactness and squared speed divided by $c^2$. L3 is the FDT source programme and each claimed divergence, the three tests below included; it is not a higher-precision version of the L1 identity. CMS describes QGP study as enabling experimental tests of quantum chromodynamics (J-020, L36–L37); the quantum face of I-4 is developed in the cone-position translation below.

A deposited energy is work, force times length; a deposition rate is power. An energy loss per length has force dimensions but remains a driven observable, not the static force in Eq. (8). Let $\mathcal{F}_{\mathrm{drv}}$ be a mechanical driving-force component, $\mathcal{H}$ a measured response, and $\chi_{\mathcal{H}}$

a causal susceptibility with response units per newton per second. Then the I-1 response form is

$$\mathcal{H}(t) = \int_{-\infty}^{t} \chi_{\mathcal{H}}(t - t')\mathcal{F}_{\mathrm{drv}}(t')\, dt'. \tag{10}$$

The susceptibility is absolutely integrable in delay and the drive bounded, or their product is absolutely integrable on the integration domain. The response vanishes with the drive and never recovers a static force by setting the drive to zero. No force of the jet in the dynamical, nonspherical medium is evaluated from the measured counts.

### 1.5 Confinement, openness and the source ledger

The helix count $n$ defines a braid class; $\Omega$ denotes its own confinement domain, $B$ its boundary data, $\chi$ orientation and $\mathfrak{t}$ confinement position. The section $u$ is a mode of the confinement operator $D$. Core I-5 specifies the eigenproblem

$$D(\Omega, B; n, \chi, \mathfrak{t})u = \omega_{\mathrm{mode}}u, \qquad m_{\mathrm{mode}}c^2 = \hbar\omega_{\mathrm{mode}}. \tag{11}$$

The operator has angular-frequency units; the mass equation contains $\hbar$, so both sides have energy units. The independent support radius $R_{\mathrm{box}}$ is geometrical data, not the Compton thread. No named pattern receives an absolute confinement address or a mass formula. The two spectral targets are kept in the status table, not used as inputs to an inferred operator.

Let $x$ be a nonnegative enhancement of the odds and $\alpha_{0,\sigma}, X_{0,\sigma}$ its initial A-1 pair. Let $\mathfrak{s}$ denote a separate orientation sign for a signed drive; $\in$ denotes membership in its stated set. The bounded update of I-6 is exactly

$$X_\sigma = X_{0,\sigma}(1 + x), \qquad \alpha_\sigma(x) = \frac{\alpha_{0,\sigma}(1 + x)}{1 + \alpha_{0,\sigma}x}, \qquad x \geq 0, \quad \mathfrak{s} \in \{-1, +1\}. \tag{12}$$

The magnitude enters $x$; polarity enters $\mathfrak{s}$. No source centrality, gap, momentum bin, angle cut or yield fraction is an application of this map. For changing gravitating energy and radius, an overdot, as in $\dot{r}_f$, denotes time differentiation. The logarithmic derivative on a fixed branch is

$$\frac{\dot{\alpha}_\sigma}{\alpha_\sigma} = \sigma\left(\frac{\dot{\mathcal{M}}}{\mathcal{M}} - \frac{\dot{r}_f}{r_f}\right). \tag{13}$$

Internal redistribution at fixed total gravitating energy and evaluation radius leaves compactness unchanged, even as the wake's distribution changes.

Let $M$ denote the rest-mass channel, $P$ pressure, $V$ volume, $T$ temperature and $a$ the radiation energy-density constant; $\sigma_{\mathrm{SB}}$ is the radiation flux constant. The collision-system I-7 ledger is

$$\mathcal{M} = M + \frac{E}{c^2} + \frac{PV}{c^2} + \frac{aT^4V}{c^2} + \cdots, \qquad a = \frac{4\sigma_{\mathrm{SB}}}{c}. \tag{14}$$

Its parent is the stress tensor of Eq. (6). Jet energy enters the energy term $E/c^2$; medium compression enters the pressure-work term $PV/c^2$; the medium's radiation enters the thermal term $aT^4V/c^2$, and the rest of its thermal energy, its heat-capacity content, enters $E/c^2$; no channel enters twice. The wake deficit redistributes these same entries; it is not another mass contribution. Gravitational binding is read from the solved geometry, not appended as

a local source channel. The record states no pressure or temperature, so these terms remain symbolic.

Every moving pattern carries a bow and a wake in the nowhere-uniform omnium. Every nonzero source carries the mass shadow, its exterior compressed-omnium response, which gravitates through $\mathcal{M}$ (0.7). Read through I-7 as compression of the substrate, it enters Eq. (14) once, through the pressure-work term that carries compression work, and never through the jet's energy term. A jet carries that response in pp as in PbPb; the core fixes no size. Its energy flow and geometry adjust together: the wake is spacetime moving. The Mach-cone-like effect that jet–medium interactions "were predicted to create" is the bow; its observation "remains elusive" (J-027–J-028, L31, L54). The wake is the source's "diffusion wake", "denoting a depletion of particles in the direction opposite to the propagating jet" (J-010, L16); calling the diffusion wake effect "the only known mechanism capable of producing such structures", the source "therefore" provides "an observation of the diffusion wake effect in the QGP" (J-175–J-177, L243–L247). The density wave and gravity's ripple are one stroke by I-2, without adding curvature at the same stress tensor and data.

The Chladni reading of P-2 is geometrical: the collision is the driver, the counted hadrons are the sand, and the stable correlation patterns are nodal architecture. Peaks mark accumulation and the depletion its complement. This reading computes no nodal pattern or density gradient of the medium, and supplies no extra equation.

### 1.6 Neighbouring patterns and the extended equivalence principle

Clocks and rulers are closed patterns whose kept time and thread length move together with the density they cross, so that $d/t$ measured on the spot stays $c$, and the I-3 optical index is the same adjustment written for a distant bookkeeper (0.1a). Let $g_{\mu\nu}$ denote the metric, $g^{\mu\nu}$ its inverse and $\partial_\nu$ a coordinate derivative. Let $\mathfrak{m}$ denote a positive dimensionless factor by which a pattern's clocks and rulers are rescaled together, $t_{\text{own}}$ the time kept on the pattern's own clock, $\mathcal{U}^\mu$ its four-velocity in that time, with $g_{\mu\nu}\mathcal{U}^\mu\mathcal{U}^\nu = -c^2$, $\nabla_{\mathcal{U}}$ the covariant rate of change along its path in $t_{\text{own}}$, and $\mathfrak{P}^{\mu\nu} = c^2 g^{\mu\nu} + \mathcal{U}^\mu\mathcal{U}^\nu$ the projector across the path. A common rescaling of $d$ and $t$ keeps every ratio of Eq. (1) and the null cone; it is the map $g_{\mu\nu} \longmapsto \mathfrak{m}^2 g_{\mu\nu}$. The connection of $\mathfrak{m}^2 g_{\mu\nu}$, with the path parametrised by $t_{\text{own}}$, gives the in-manuscript identity: a pattern free of every force in $\mathfrak{m}^2 g_{\mu\nu}$ is read by the bookkeeper, on the unrescaled metric, with

$$\nabla_{\mathcal{U}}\mathcal{U}^\mu = -\mathfrak{P}^{\mu\nu}\partial_\nu \ln \mathfrak{m}. \tag{15}$$

On its own clocks and rulers the pattern meets nothing; the bookkeeper reads an acceleration toward smaller $\mathfrak{m}$, where the pattern's clocks run slower. By P-4 the density cone is the local light cone, and refraction, lensing and horizon formation are one phenomenon. Moving at constant local speed through a density gradient is therefore gravity, and no reading on the pattern's own clocks and rulers separates the two: this is the extended equivalence principle.

Let $\mathfrak{e}_{\text{ray}}$ denote a stroke's unit direction, $\ell_{\text{geo}}$ its geometric arclength and $\nabla_\perp$ the gradient across it. The Fermat law of a positive index $\mathfrak{n}$ bends the stroke by $d\mathfrak{e}_{\text{ray}}/d\ell_{\text{geo}} = \nabla_\perp \ln \mathfrak{n}$, which the bookkeeper reads, at the stroke's coordinate speed $c/\mathfrak{n}$, as the transverse acceleration $(c/\mathfrak{n})^2 \nabla_\perp \ln \mathfrak{n}$. The isotropic weak-field index of I-3, $\mathfrak{n}_{\text{iso}} = 1 + \alpha_{\text{ext}} + O(\alpha_{\text{ext}}^2)$ in its own chart, gives $c^2 R_s/r_f^2 = 2G\mathcal{M}/r_f^2$ at leading order: twice the Newtonian value, the factor of two of I-3, from the principle.

The value of $\ln \mathfrak{m}$ at a point is invisible on the pattern's own clocks and rulers; its gradient is the free fall of Eq. (15), removed by riding with the pattern; its second derivatives survive every such choice and are read only by comparing neighbouring patterns. Let $\mathcal{J}$ denote the separation of two neighbouring paths, qualified in and out for its components in and across the plane of a stroke's pass, with $\ell_{\mathrm{opt}}$ the optical path length of Eq. (5) along the stroke. Let $\mathfrak{K}_{\mathrm{in}}$ and $\mathfrak{K}_{\mathrm{out}}$ denote the sectional curvatures of that optical metric on planes containing the radial direction and on planes tangent to its spheres. The second variation of the Fermat law is the Jacobi equation of the optical metric; on the exterior branch, exactly,

$$\begin{aligned} \frac{d^2 \mathcal{J}_{\mathrm{in}}}{d\ell_{\mathrm{opt}}^2} &= -\mathfrak{K}_{\mathrm{in}} \mathcal{J}_{\mathrm{in}}, \qquad \mathfrak{K}_{\mathrm{in}} = -\frac{\alpha_{\mathrm{ext}}}{r_f^2}\left(1 - \frac{3\alpha_{\mathrm{ext}}}{4}\right), \\ \mathfrak{K}_{\mathrm{out}} &= \frac{2\alpha_{\mathrm{ext}}}{r_f^2}\left(1 - \frac{9\alpha_{\mathrm{ext}}}{8}\right). \end{aligned} \tag{16}$$

The plane of the pass is totally geodesic, so the first line holds along the whole stroke and neighbours in that plane separate. Across it the separation is an eigen-direction whose curvature at closest approach is $\mathfrak{K}_{\mathrm{out}}$, and neighbours converge. Stretching in the plane and compression across it are the tides of general relativity read on strokes; at closest approach their trace-free part is the null shear of Eq. (6) at leading order. Let $r_{\mathrm{ca}}$ denote the closest-approach radius and $\theta_{\mathrm{def}}$ the deflection of one stroke against its undeflected neighbour. By the Gauss–Bonnet theorem $\theta_{\mathrm{def}}$ is minus the integral of $\mathfrak{K}_{\mathrm{in}}$ over the optical area between the stroke and the far field on the side away from the source, which is $2R_s/r_{\mathrm{ca}} = 4G\mathcal{M}/(r_{\mathrm{ca}}c^2)$ at leading order, the I-3 deflection: the bend of one stroke is the curvature it encloses with a reference path in the far field.

For neighbouring patterns at rest or falling radially on the exterior branch, on their own clocks and in the frame of the radial and the two transverse directions, exactly,

$$\nabla_{\mathcal{U}} \nabla_{\mathcal{U}} \mathcal{J} = \frac{c^2 \alpha_{\mathrm{ext}}}{2 r_f^2}\, \mathrm{diag}(2, -1, -1)\, \mathcal{J}. \tag{17}$$

This is the tidal stretching along the radius and compression across it: the curvature ratio times the squared curvature $1/r_f^2$ of the field circle (0.3), with zero trace on the exterior and the finite radial value $c^2/R_s^2$ as $r_f \to R_s^+$. For null strokes the trace-free part per unit optical length squared is twice that of Eq. (17) per unit $(c t_{\mathrm{own}})^2$ at leading order, the factor of two again. With Eq. (3), $R_s = \mathcal{M}c^2/(2F_{\max})$, so a pattern's own compactness needs no numerical gravitational constant, $\alpha_{\mathrm{ext}} = \mathcal{M}c^2/(2F_{\max} r_f)$. For the source's subleading threshold, $p_{\mathrm{T}}^{\mathrm{j}2} > 50\,\mathrm{GeV}$ (J-072), the jet's own $\mathcal{M}c^2$ exceeds 50 GeV, and at a declared $r_f$ of 1 fm (U-41, U-42) $\alpha_{\mathrm{ext}}$ exceeds $1.3 \times 10^{-37}$. That small number is the correct gravitational reading of the jet's own deviation, as the small forces of I-1 are correct readings, and the wake's curvature remains that of Eq. (6).

For the medium, let $E/V$ and $P$ denote its energy density and pressure. For a stress tensor of perfect form at zero net conserved count, the conservation that Eq. (6) imposes on its source, projected across the path, and the Gibbs–Duhem relation give

$$\begin{aligned} \left(\frac{E}{V} + P\right) \nabla_{\mathcal{U}} \mathcal{U}^\mu &= -\mathfrak{P}^{\mu\nu} \partial_\nu P, \qquad dP = \left(\frac{E}{V} + P\right) \frac{dT}{T}, \\ \nabla_{\mathcal{U}} \mathcal{U}^\mu &= -\mathfrak{P}^{\mu\nu} \partial_\nu \ln T \end{aligned} \tag{18}$$

for every equation of state, which is Eq. (15) with $\mathfrak{m}$ proportional to $T$. The medium's patterns are free of every force in $T^2 g_{\mu\nu}$: the temperature of Eq. (14) is their clock factor, the entries $E$ and $PV$ per volume are their inertia, and the pressure force is the bookkeeper's reading of their common rescaling. The jet's deposition carries that structure with the jet, the medium's patterns free-fall through it, and the second derivatives of $\ln T$ set their deviation at the medium's own strength, which the record does not print. The medium's strokes bend instead by the narrowing of the density cone (P-4), which no conformal factor supplies; the bow is the caustic where strokes emitted along the jet's path cross, and its opening needs the constitutive derivation of that aperture (O-8). One pattern carries no reading of the wake: the wake is the deviation of neighbouring patterns.

## 2 System chart table

Let $\rho_{\mathrm{loc}}(r_f)$ denote local density, $\bar{\rho}$ the enclosed mean density and $V_{r_f}$ its volume. Let $w$ denote a nonnegative probe weight and $f$ a sampled local quantity; $\langle f \rangle_w$ is its normalized weighted mean. For an integrable numerator and positive finite denominator,

$$\bar{\rho} = \frac{\mathcal{M}}{V_{r_f}}, \qquad \langle f \rangle_w = \frac{\int w f \, dV}{\int w \, dV}. \tag{19}$$

The mean-density proxy on the interior branch is not the local profile. The local density is largest at the skin and dilute at the centre, for every construction with an inside (0.6). The radius $R_{\mathrm{room}}$ is a measured interior support radius, independent of the exterior gravitational radius; no such nuclear radius is printed by CMS in this record.

| Referent | Chart and entry point | Observation-region rule |
|---|---|---|
| Whole collision | Exterior A-1 from total $\mathcal{M}$ and declared $r_f$; gap odds in Eq. (4) | No numerical compactness without these inputs; laboratory far-field reading. |
| Lead nucleus | Exterior fold $R_s$; interior skin fold with independent $R_{\mathrm{room}}$ | Two charts, no transfer map or rescaling of the laboratory radius. |
| Nuclear pinning skin | $\alpha_{\mathrm{core}} \to 1^-$ in its own interior chart | Saturation is a separate channel from exterior compactness. |
| Charged-hadron depletion | Count-weighted projection of final-state closed patterns | Per dijet pair and per radian, near side, long-range background subtracted; no further interior weight given. |
| Centrality classes | HF transverse-energy-sum selection | A percentile of the inelastic hadronic cross section, not an interior weight or density value. |
| Jet bow and wake | Driven response, Eq. (10); source entries counted once | No count or significance is a force; no medium sound speed supplied. |
| Neighbouring patterns | Index geometry of each window: Eq. (15) for patterns, Eq. (5) for strokes | One pattern carries no reading; the deviation of neighbours, Eqs. (16)–(18), is the reading. |

The centrality, momentum and gap dependence all inherit the same final-state charged-hadron count weight. They do not reconstruct a local radial profile. The source's HF selection and its normalization are J-042–J-043, J-056–J-059 and J-089; the near-side and background selections are J-124–J-128.

## 3 Observation map

The source reports $E_{\mathrm{NN}} = 5.02\,\mathrm{TeV}$, $0.66\,\mathrm{nb}^{-1}$ PbPb integrated luminosity in 2018 and $299\,\mathrm{pb}^{-1}$ pp integrated luminosity in 2017 (J-012, J-034–J-036). The recorded cuts and estimator define the common input dictionary. No numerical yield, bin covariance or exact significance is printed. No yield is inferred from a figure or from a cited external data table.

| Source section or locus | Observation or definition | Epistemic class |
|---|---|---|
| Jet selection, L93–L112 | Leading $p_{\mathrm{T}}^{\mathrm{j1}} > 130\,\mathrm{GeV}$, $\lvert\eta^{\mathrm{j1}}\rvert < 1.0$; subleading $p_{\mathrm{T}}^{\mathrm{j2}} > 50\,\mathrm{GeV}$, $\lvert\eta^{\mathrm{j2}}\rvert < 2.0$; $\Delta\Phi_{\mathrm{j1,j2}} > 5\pi/6$ (J-071–J-074). | Reported cuts. |
| Gap classes, L108–L112 | Small $\lvert\Delta\eta_{\mathrm{j1,j2}}\rvert < 0.5$; large intervals $(0.5, 1.0)$, $(1.0, 1.5)$, $(1.5, 2.0)$ (J-075–J-077). | Reported cuts. |
| Tracks, L114–L116 | $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$ or $2 < p_{\mathrm{T}}^{\mathrm{ch}} < 4\,\mathrm{GeV}$, $\lvert\eta\rvert < 2.4$ (J-078–J-079). | Reported cuts. |
| Centrality, L128–L133 | Percentiles 0–30%, 30–50%, 50–80% from HF transverse energy (J-056–J-059). | Reported selection. |
| Correlation, L132–L159 | Leading pseudorapidity ordered above subleading; acceptance and track corrections; per-pair normalization (J-087–J-099). | Corrected estimator. |
| Near side, L183–L191 | $\lvert\Delta\Phi^{\mathrm{ch,j1}}\rvert < \pi/2$; background from $1.5 < \Delta\eta^{\mathrm{ch,j1}} < 2.5$ removed separately; asymmetric minus symmetric (J-124–J-128). | Reported definition. |

Let $\mathcal{D}$ denote the near-side asymmetric-minus-symmetric contrast, with a system label for PbPb or pp; it has the same yield-density units as $\mathcal{R}$. Let $\mathcal{Y}$ denote its integral, in yield per pair per radian. Source J-128, J-144 and J-169 define

$$\begin{aligned} \mathcal{D}_{\mathrm{system}} &= \mathcal{R}_{\mathrm{asym,system}} - \mathcal{R}_{\mathrm{sym,system}}, \\ \Delta\mathcal{D} &= \mathcal{D}_{\mathrm{PbPb}} - \mathcal{D}_{\mathrm{pp}}, \\ \mathcal{Y}_{\mathrm{PbPb-pp}} &= \int_{-2.4}^{-0.6} \Delta\mathcal{D}\, d(\Delta\eta^{\mathrm{ch,j1}}). \end{aligned} \tag{20}$$

The first subtraction exposes the displaced wake; the second compares systems. Reapplying these definitions to the identical corrected inputs is the identity map of the data. It corroborates no invariant and is not an I-1 force calculation. The integrated units retain the per-radian normalization.

For small gaps the source states that "the diffusion wakes are hidden by the jet peaks, and no wake signal is visible" (J-121, L175–L176); its omitted corrections are those that "cancel when subtracting" large- and small-gap distributions (J-096, L154), and its simulation without medium effects shows no underlying-event variation with the subleading jet location, "as expected" (J-093–J-094). Eq. (20) is therefore a mixed second difference over two neighbour coordinates, gap and system: what neighbouring selections share cancels, as the common free fall of a pair of patterns cancels in Eq. (17), and the relative part remains. One dijet or one gap class carries no wake reading.

| Source locus | Observation or comparison | Epistemic class |
|---|---|---|
| Figure 1, L187–L194 | In $(1.0, 1.5)$ and $-2.4 < \Delta\eta^{\mathrm{ch,j1}} < -0.6$, the pp distributions are approximately flat, with minimal deviation from zero; PbPb collisions show a pronounced dip, most prominent at low $p_{\mathrm{T}}^{\mathrm{ch}}$ (J-129–J-135). | Measured, qualitative. |
| L193–L197 | The strength of the depletion increases from peripheral (50–80%) to central (0–30%) collisions; the 0–30% PbPb data for $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$ lie significantly below the pp reference (J-134–J-137). | Reported comparison. |
| Figure 3, L233–L240 | Integral over $-2.4 < \Delta\eta^{\mathrm{ch,j1}} < -0.6$, for $(1.0, 1.5)$ and $(1.5, 2.0)$; central 1–2 GeV deviates from zero by more than five standard deviations (J-168–J-173). | Reported bound, chi-squared test. |
| L193–L203 | Unequal away-side jet-peak positions preclude direct subtraction there (J-138–J-140). | Reported limitation. |
| L202–L229 | In the 0–30% PbPb-minus-pp difference, for $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$ the mean position of the observed depletion shifts toward negative $\Delta\eta^{\mathrm{ch,j1}}$ as the gap selection increases from $(0.5, 1.0)$ to $(1.5, 2.0)$; medium-induced jet peak modifications and out-of-cone radiation can interfere with the diffusion wake signal (J-144, J-147–J-150). | Observed trend; reported interpretation. |
| L242–L247 | The source calls the diffusion wake effect "the only known mechanism capable of producing such structures" and "therefore" provides "an observation of the diffusion wake effect in the QGP" (J-175–J-177). | Exclusion interpretation. |

These measurements are accepted at their stated precision: a qualitative shape and a lower significance bound, without an invented depletion depth. The same charged-hadron count weight applies to every interior reading in the table. The I-2 reading identifies the stroke and P-2 identifies the nodal architecture; neither adds a numerical agreement beyond the identity map. The independent tests are the polarization, charge and pp observables below: a positive pp excess above the three-standard-deviation threshold in either class falsifies the depleted sign of T-3, and a null in any of them bounds the response it tests. I-3 corroboration would require independently specified stress–energy and curvature with uncertainties; these yields provide no such independent pair.

The residual underlying-event level uncertainty is 0.01–0.04, pair acceptance 0–0.01, and tracking 0–0.025, in the source's yield units (J-103, J-107, J-111). CMS conservatively applies the maximum observed tracking variation to all $\Delta\eta^{\mathrm{ch,j1}}$ bins and adds sources in quadrature (J-102, J-115). The record states no inter-bin covariance for the other terms and no covariance used in its chi-squared significance. Thus its bound cannot be independently reconstructed here; no substitute test statistic is attributed to CMS.

| Source model comparison | Reading with the source's precision |
|---|---|
| PYTHIA, L236–L238 | No $\Delta\eta^{\mathrm{ch,j1}}$ dependence, "as expected in the absence of jet quenching" (J-159–J-160); a simulation. |
| HYBRID, L239–L249 | In its fully incoherent energy loss scenario the model overpredicts the observed depletion for $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$; the source says the subleading wake "should reduce" the symmetric near-side peak (J-161–J-167); a model mechanism. |
| CoLBT-hydro, L220–L231 | The calculation uses 0–10% centrality against 0–30% data; comparison "is impaired" by that difference, and the dip position "appears to be captured" by both models (J-151–J-156). |
| Figure 2 caption, L285–L288 | Data statistical and systematic uncertainties are distinct; model bands show statistical uncertainties only (J-184–J-186). |

The computed bands are not measurements of a density profile or force. Their common-input agreement has no additional FDT precision attached. I-3 containment returns the same curvature from the same stress tensor and data, and I-4 L2 applies at the stated weak-field, slow-motion order; neither asserts that a phenomenological yield model is derived by a coordinate change. An independently measured violation of either containment relation would falsify the corresponding invariant in its stated domain.

## 4 Cone-position translation

Quantum mechanics is helical geodesics on truncated density cones (I-4). The universe is deterministic; there are no probability amplitudes (I-4). Let $\psi$ be the complex projection of a helical mode, $A$ its transverse amplitude, $\phi$ its phase and $i$ the imaginary unit. The phase origin is $\phi_0$ and the helical phase $\phi_h(t)$. Let $\mathfrak{x}, \mathfrak{y}, \mathsf{z}$ be cylindrical-lift Cartesian coordinates and $v_{\mathrm{ax}}$ the axial transport speed. Then

$$\begin{aligned} \psi &= A\exp(i\phi_h), \qquad \phi_h(t) = \phi_0 - \omega t, \\ \mathfrak{x} &= A\cos\phi_h, \qquad \mathfrak{y} = A\sin\phi_h, \qquad \mathsf{z} = v_{\mathrm{ax}} t. \end{aligned} \tag{21}$$

Here $A$ has length units for the geometric lift; the exponent is dimensionless. Ignorance of the phase origin at observation is superposition. Let $e_1, e_2$ be orthonormal Clifford basis directions and $\mathbf{B} = e_1 e_2$ their unit oriented bivector. The Clifford form is

$$i \longleftrightarrow \mathbf{B}, \qquad \exp(\mathbf{B}\phi_h) = \cos\phi_h + \mathbf{B}\sin\phi_h. \tag{22}$$

This is a single-angle phase rotor, distinct from a half-angle spinor rotor. It is a phase-axis lift with fixed field identity, not a change of strand count or a density token promoted to a multivector. The macroscopic jet correlations do not measure this phase.

The counted hadrons are closed braid configurations with no intrinsic mass, charge or colour. Quarks are sub-braid strands; their number equals the helix count. Colour is the SU(3) of the internal nodal surface of $n = 3$ braids (I-5). Confinement is topological closure, not a separate force. Mass follows confinement position and the spectrum in Eq. (11). The core pairs electron and proton at the same helix count while their confinement positions differ; it assigns no helix count to the lambda hyperon. The proton–pion and antiproton–pion final states used below are consensus reconstructions, not an asserted FDT transition. No same-count or changed-count transition for that configuration is inferred.

| Observable or region | Cone-position translation |
|---|---|
| Exterior collision chart | Curvature ratio from total gravitating energy and a declared radius, not from a yield fraction. |
| Interior nuclear skin | Saturated limiting boundary of its own chart, with a dilute centre and independently specified support. |
| Hadron count | Counts of closed patterns after reconstruction; no absolute mode address. |
| Jet energy deposited internally | Source-channel relabelling; fixed total energy and radius leave compactness fixed. |
| Bow geometry | The Mach-cone-like effect jet–medium interactions "were predicted to create" (J-027, L54); the aperture candidate supplies no sound-cone angle. |
| Wake correlations | Count-weighted driven geometry and nodal architecture; no local medium profile extracted. |
| Pattern crossing a density gradient | Constant local speed on its own clocks and rulers; the bookkeeper's acceleration of Eq. (15); gravity by the extended equivalence principle. |

# 5 Predictions

The following three blocks add a source or an observable dictionary to the convergence calculation. They are conditional by construction. Existence means a nonzero projected response with no fixed amplitude; a finite null result bounds that response, not an unspecified lower bound. For each block the code $\mathcal{E}$ denotes existence, with FDT code +1, the comparator code 0, and code threshold 1 (U-10). These dimensionless codes are not experimental signal-to-noise ratios. Experimental decisions separately use $\mathfrak{z} = 3$ times the standard deviation of the block's own statistic, with covariance (U-09).

Each block compares neighbouring patterns, arranged so that the comparator's deviation cancels or vanishes. In T-1 the neighbours are one configuration in opposite handedness in the same wake, and spin–vorticity polarization acts on both alike, as a common free fall acts on both members of a pair in Eq. (17). In T-2 they are opposite charges across two annuli, and the charge-even depletion acts on both alike. In T-3 they are the two gap selections in pp, where the comparator has no medium response; the moving source's mass shadow is a density gradient that the pp patterns cross, and by the extended equivalence principle it deviates them. The trace of the deviation reads density, the measured depletion and T-3; its antisymmetric part reads circulation and its magnetic projection, T-1 and T-2; its trace-free symmetric part, the tidal stretch and compression, has no observable map in the record and no additional source at fixed stress tensor.

## 5.1 A response kernel for the circulation tests

The source says that energetic quarks and gluons deposit energy and momentum before hadronizing (J-007–J-009, L13–L15). In consensus fluid dynamics, flow along an axis with transverse variation has curl circulating about that axis. Denote the corresponding mechanical circulation by $\mathfrak{w}$. P-5 reads mechanical vorticity and magnetic moments as different grade projections of one density-gradient topography. The field $\mathcal{B}_{\mathrm{tor}}$ is this helical-gradient projection, not an independent entity. Its direction reverses with the jet's direction.

To specify the response form, let $\chi_{\mathfrak{w}}$ be circulation susceptibility to the I-1 mechanical drive of Eq. (10), and let $\chi_{\mathcal{B}}$ map that circulation to its magnetic projection. Their units are circulation per newton per second and tesla per circulation per second. Both kernels are causal and absolutely integrable; let $\mathcal{K}$ denote their convolution, with units tesla per newton per second. Fubini's theorem then gives the in-manuscript derivation

$$\begin{aligned}
\mathfrak{w}(t) &= \int_{-\infty}^{t} \chi_{\mathfrak{w}}(t-t')\mathcal{F}_{\mathrm{drv}}(t')\,dt', \\
\mathcal{B}_{\mathrm{tor}}(t) &= \int_{-\infty}^{t} \chi_{\mathcal{B}}(t-t')\mathfrak{w}(t')\,dt', \\
\mathcal{K}(t) &= \int_{0}^{t} \chi_{\mathcal{B}}(t-t')\chi_{\mathfrak{w}}(t')\,dt', \\
\mathcal{B}_{\mathrm{tor}}(t) &= \int_{-\infty}^{t} \mathcal{K}(t-t')\mathcal{F}_{\mathrm{drv}}(t')\,dt'.
\end{aligned} \tag{23}$$

I-1 supplies the mechanical input, I-2 its single carrier, and P-5 the projection dictionary. This is a constitutive response family, not a numerical kernel derived from the charge-summed yield. The core fixes neither its sign nor its normalization. The existence tests condition on nonzero circulation, nonzero projected kernel overlap and an acceptance-weighted annulus that does not cancel the projection. Ablation sets the additional projection kernel to zero while preserving the mechanical circulation and the conventional controls. No instantaneous sensitivity times a span is used as an amplitude. For a general positive observable $\mathcal{O}$ and a dimensionless path parameter $\mathfrak{u}$, the sensitivity $\mathsf{k}_{\mathcal{O}}$ is its logarithmic derivative along that path, so the amplitude condition is

$$\Delta \ln \mathcal{O} = \int_{\mathfrak{u}_{\mathrm{in}}}^{\mathfrak{u}_{\mathrm{out}}} \mathsf{k}_{\mathcal{O}}(\mathfrak{u})\,d\mathfrak{u}. \tag{24}$$

This identity does not supply an unmeasured kernel value.

### 5.2 T-1: toroidal polarization at zero jet charge

DC-1 discriminates SA-38: a model of the charge-summed wake does not specify the opposite-species toroidal-polarization intercept. The referent is the lambda hyperon and antihyperon, consensus species names for one configuration in opposite handedness (P-4), compared at the same helix count, whose value the core does not fix, and reconstructed in the CMS dijet selection of the observation table, with absolute dijet gap from 1.0 to 2.0, in 0–30% PbPb and pp at 5.02 TeV (J-071–J-077, J-130, J-170). The polarization decision uses PbPb; the pp statistic is reported beside it.

Use a beam-direction boost to the subleading jet's rapidity, where that jet is transverse. Under this boost a massive hyperon's rapidity shifts by the boost rapidity, while its pseudorapidity shifts by a momentum-dependent amount. The wake centre is opposite the jet at its rapidity. Let $\mathfrak{d}_{\mathrm{w}}$ be angular distance from that centre in rapidity difference and azimuth difference. Fix the hyperon annulus $0.2 < \mathfrak{d}_{\mathrm{w}} < 0.6$ before reading the data (U-30). The toroidal unit direction $\mathfrak{e}_{\mathrm{tor}}$ is the jet's unit direction $\mathfrak{e}_{\mathrm{jet}}$ crossed with the hyperon's $\mathfrak{e}_{\mathrm{hyp}}$, normalized when the cross product is nonzero. It winds once around the wake centre. Define $\Theta^{*}$ as the angle of the consensus decay proton, or decay antiproton, relative to this axis in

the reconstructed hyperon rest frame. The labels hyp and $\overline{\text{hyp}}$ distinguish the hyperon and antihyperon in species subscripts. The species decay parameter is $\mathfrak{a}_{\text{dec}}$, and $\mathcal{P}^{\text{tor}}$ denotes the corresponding polarization. Consensus angular reconstruction, with normalized angular density $\mathfrak{p}$, gives

$$\mathfrak{e}_{\text{tor}} = \frac{\mathfrak{e}_{\text{jet}} \times \mathfrak{e}_{\text{hyp}}}{\|\mathfrak{e}_{\text{jet}} \times \mathfrak{e}_{\text{hyp}}\|}, \quad \mathfrak{p}(\cos\Theta^*) = \frac{1 + \mathfrak{a}_{\text{dec}}\mathcal{P}^{\text{tor}}\cos\Theta^*}{2}. \tag{25}$$

Let $\mathcal{Q}_{\text{jet}}$ be the net subleading-track charge after subtracting the underlying-event charge in an equal cone. Let $\mathcal{I}$ denote an intercept functional: a weighted straight-line fit of a species polarization difference against this charge, with the regressor's error distribution measured anew from random cones. The weights and charge mean are retained; a nonzero charge mean cannot be silently centred without transforming the intercept back to zero charge. Let $\mathcal{C}_{\text{P}}$ denote the jet-axis minus mixed-axis intercept, where the mixed subleading axis comes from an independent event in the same centrality class. Define its covariance matrix $\mathcal{V}_{\text{P}}$ and contrast coefficient vector $\mathfrak{l}_{\text{P}}$. The superscript $\mathsf{T}$ denotes matrix transpose. The defining condition and its linear solution are

$$\begin{aligned} \mathcal{C}_{\text{P}} &= \mathcal{I}[\mathcal{P}^{\text{tor}}_{\text{hyp}} - \mathcal{P}^{\text{tor}}_{\overline{\text{hyp}}}]_{\text{jet}} - \mathcal{I}[\mathcal{P}^{\text{tor}}_{\text{hyp}} - \mathcal{P}^{\text{tor}}_{\overline{\text{hyp}}}]_{\text{mix}}, \\ \mathrm{sd}^2(\mathcal{C}_{\text{P}}) &= \mathfrak{l}_{\text{P}}^{\mathsf{T}}\mathcal{V}_{\text{P}}\mathfrak{l}_{\text{P}}, \qquad |\mathcal{C}_{\text{P}}| > \mathfrak{z}\,\mathrm{sd}(\mathcal{C}_{\text{P}}). \end{aligned} \tag{26}$$

For a known linear design matrix $\mathsf{J}$ and data vector $\mathfrak{v}$ with covariance $\mathcal{V}$, the fitted intercept and slope vector $\mathfrak{b}$ is the solution

$$\mathfrak{b} = (\mathsf{J}^{\mathsf{T}}\mathcal{V}^{-1}\mathsf{J})^{-1}\mathsf{J}^{\mathsf{T}}\mathcal{V}^{-1}\mathfrak{v}. \tag{27}$$

The design has full column rank and the covariance is positive definite. With noisy charge the calibrated latent-charge likelihood replaces a naive measured-charge design. Thus a term proportional to true jet charge has zero population intercept after resolution correction.

The comparator is zero after the following consensus controls, which are not CMS measurements in this record. Spin–vorticity coupling polarizes the two species alike and cancels in their difference. Their opposite magnetic moments make the circulation's magnetic projection species odd. The jet's own charge field vanishes at the zero-charge intercept. A uniform field cancels over the full winding annulus after acceptance correction. The species detector offset cancels against the mixed axis; its mismatch remains in the covariance. Three surviving C-odd terms are retained: baryonic spin-Hall polarization, the magnetic field of bulk net-charge flow, and transverse fragmentation polarization [3]. The first two are bounded with bulk proton–antiproton and charge asymmetries; the third with the in-jet hyperon-yield asymmetry and mean jet charge. The annulus suppresses, but does not assume away, that fragmentation contribution.

Let $\mathfrak{h}$ be a signed raw projected response and $\mathfrak{h}_0$ a positive reference with the same units. The bounded, unnormalised response is $\mathcal{S}$; it retains the zero-drive value:

$$\mathcal{S} = \frac{\mathfrak{h}}{\mathfrak{h}_0 + |\mathfrak{h}|}, \qquad \mathfrak{h} = 0 \implies \mathcal{S} = 0. \tag{28}$$

A nonzero kernel integral in Eq. (23) changes the opposite species in opposite senses; zero additional projection gives the comparator in Eq. (26). The sign is not specified. This is an

existence prediction: $\mathcal{E}_{\mathrm{FDT}} = +1$, $\mathcal{E}_{\mathrm{ablated}} = 0$, with code threshold 1. A null measurement limits the projected response; a claimed nonzero response must survive every stated C-odd bound and both charge-resolution and decay-parameter calibrations.

| Co-varying channel | Sign and surviving term |
|---|---|
| Mechanical vorticity, species even | Same species polarization; ideal difference zero. |
| Helical-gradient projection | Opposite species; either sign; reverses with the jet-defined axis. |
| Jet-charge field | Odd in charge; zero corrected intercept. |
| Uniform field and detector offset | Winding average and mixed-axis cancellation, with mismatch covariance. |
| Spin-Hall, bulk-current, fragmentation | Species odd; independent bounds survive both subtractions. |

### 5.3 T-2: charge separation across fixed annuli

DC-2 discriminates SA-30: the measured charge-summed correlation does not fix a charge-odd radial contrast. Use the charged hadrons in the 0–30% PbPb large-gap events of the classes $(1.0, 1.5)$ and $(1.5, 2.0)$, pooled into one sample and one decision, with $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$ (J-130, J-169–J-171). The small-gap sample and pp are excluded from this decision; the pp contrast is reported as a separate control. The End Matter supplies charge-summed angular distributions only for the small gap and the first of these large-gap classes (J-187–J-193).

Centre the annuli at the subleading pseudorapidity and opposite azimuth. For these track annuli $\mathfrak{d}_{\mathrm{w}}$ uses laboratory pseudorapidity rather than hyperon rapidity. Fix the inner interval $(0.2, 0.4)$ and outer interval $(0.4, 0.6)$ (U-30). The strict outer bound is clear of the leading cone of distance parameter 0.4 at the minimum gap 1.0; its boundary touches but its interior does not overlap that cone (J-066, J-130). The radial senses are assigned in the same beam-boosted frame as above. The consensus force on a moving charge in a magnetic field separates opposite charges radially when their motion is opposite the jet. A hadron's charge sign is the nested-helix sign of I-5, not its helix count; each charge-conjugate pair is one configuration in opposite handedness (P-4), compared at the same helix count. P-5 supplies the additional field projection, with neither sign nor magnitude fixed by the core.

Let $\mathcal{N}_{\mathrm{in}}^{\pm}$ and $\mathcal{N}_{\mathrm{out}}^{\pm}$ be corrected charge-separated yields in the fixed annuli. Let $\mathcal{A}_{\mathrm{odd}}$ and $\mathcal{A}_{\mathrm{even}}$ denote their odd and even contrasts at the resolution-corrected zero-charge intercept. Let $\mathcal{N}_{\mathrm{bulk}}$ and $\mathcal{Q}_{\mathrm{bulk}}$ be the total and signed counts in a large bulk control region of the same events. The bulk asymmetry $\mathfrak{q}_{\mathrm{bulk}}$ is their population ratio, never a noisy ratio in the decision. The product statistic $\mathcal{C}_{\mathrm{A}}$ is defined by

$$\begin{aligned} \mathcal{A}_{\mathrm{odd}} &= \mathcal{I}[(\mathcal{N}_{\mathrm{in}}^{+} - \mathcal{N}_{\mathrm{out}}^{+}) - (\mathcal{N}_{\mathrm{in}}^{-} - \mathcal{N}_{\mathrm{out}}^{-})], \\ \mathcal{A}_{\mathrm{even}} &= \mathcal{I}[(\mathcal{N}_{\mathrm{in}}^{+} - \mathcal{N}_{\mathrm{out}}^{+}) + (\mathcal{N}_{\mathrm{in}}^{-} - \mathcal{N}_{\mathrm{out}}^{-})], \\ \mathcal{C}_{\mathrm{A}} &= \mathcal{N}_{\mathrm{bulk}}\mathcal{A}_{\mathrm{odd}} - \mathcal{Q}_{\mathrm{bulk}}\mathcal{A}_{\mathrm{even}}, \qquad |\mathcal{C}_{\mathrm{A}}| > \mathfrak{z}\,\mathrm{sd}(\mathcal{C}_{\mathrm{A}}). \end{aligned} \tag{29}$$

With the same charge-dependent acceptance prefactor in both annuli, the bulk term removes charge imbalance and detector asymmetry from every charge-even response if and only if both annuli preserve the bulk's positive-to-negative ratio; for one response the population product vanishes if and only if the two charges' inner-minus-outer contrasts keep that ratio. A charge-even depletion alone then gives exactly zero population product. In consensus

physics, charge/baryon diffusion, mass-dependent response and axial bulk-charge current break that condition. Their bounds scale with independently measured bulk charge and proton–antiproton asymmetries. In consensus physics, leading-jet fragments have a nonzero mean charge from the beams' valence quarks. Their leakage beyond the leading jet's cone survives the subleading intercept. Its bound uses the leading jet's charge and measured outer-annulus yield.

For a compact exact covariance expression, collect the two bulk counts and two fitted contrasts into their joint four-vector, and let $\mathfrak{H}$ be the constant Hessian of their bilinear product. Let $\mathfrak{g}$ be its gradient at the population mean and $\mathcal{V}$ the joint covariance. For a joint Gaussian vector the product variance is exactly

$$\mathrm{sd}^2(\mathcal{C}_\mathrm{A}) = \mathfrak{g}^\mathsf{T}\mathcal{V}\mathfrak{g} + \frac{1}{2}\,\mathrm{tr}(\mathfrak{H}\mathcal{V}\mathfrak{H}\mathcal{V}). \tag{30}$$

For independent Poisson bulk counts and annulus data, the equivalent second-moment product formula is exact without a Gaussian approximation. The numerical experiment below uses that formula, including bulk-count fluctuations, charge-calibration error and the bounded systematic terms.

The projected kernel nonzero condition of Eq. (23) and a noncancelling charge transport across the boundary give the existence prediction $\mathcal{E}_\mathrm{FDT} = +1$ versus $\mathcal{E}_\mathrm{ablated} = 0$, with code threshold 1. The comparator is zero after the stated bounds, not zero by charge summation. The kernel's sign remains free; charge reversal reverses the radial response, while charge-even depletion changes both yields alike. This explicit single-channel scope concerns the additional circulation projection; the charge/baryon and leading-leakage channels remain in the covariance and are not counted as its signal.

### 5.4 T-3: the pp wake integral

DC-7 discriminates SA-32 by testing the mass shadow carried by a pp jet, an additional source under core 0.7. The referent is the CMS pp dijet selection at 5.02 TeV, with $1 < p_\mathrm{T}^\mathrm{ch} < 2\,\mathrm{GeV}$, for each of the two large-gap classes $(1.0, 1.5)$ and $(1.5, 2.0)$ (J-071–J-077, J-169–J-170). The core fixes no size of this depletion. The source's pp distributions in the first class are "approximately flat, with minimal deviation from zero" (J-132, L190–L191), accepted in full. That statement bounds the depletion there; it is not an exact zero and states nothing about a pp-only integral in the second class.

Let $\mathcal{Y}_{\mathrm{pp},j}$ denote the pp integral for gap class $j$, where $j$ here and below indexes the two gap classes. Let $\mathsf{w}_j$ be its vector of bin widths and $\mathcal{V}_j$ the covariance of its yield-density bins. Let $\mathsf{u}_{j,\mathsf{r}}$ denote the signed one-standard-deviation bin shift due to systematic source $\mathsf{r}$, with $\mathsf{r}$ a label for residual level, pair acceptance or tracking. The statistical covariance is $\mathcal{V}_{j,\mathrm{stat}}$. The defining condition and decision are

$$\begin{aligned} \mathcal{Y}_{\mathrm{pp},j} &= \int_{-2.4}^{-0.6} \mathcal{D}_{\mathrm{pp},j}\, d(\Delta\eta^{\mathrm{ch,j1}}), \\ \mathcal{V}_j &= \mathcal{V}_{j,\mathrm{stat}} + \sum_{\mathsf{r}} \mathsf{u}_{j,\mathsf{r}}\mathsf{u}_{j,\mathsf{r}}^\mathsf{T}, \\ \mathrm{sd}^2(\mathcal{Y}_{\mathrm{pp},j}) = \mathsf{w}_j^\mathsf{T}\mathcal{V}_j\mathsf{w}_j, &\qquad \mathcal{Y}_{\mathrm{pp},j} < -\mathfrak{z}\,\mathrm{sd}(\mathcal{Y}_{\mathrm{pp},j}) \quad \text{for both classes.} \end{aligned} \tag{31}$$

Each systematic is fully correlated across bins within its class; sources add in quadrature. This is a conservative bin-correlation choice where CMS supplies no covariance, and retains its

all-bin tracking shift. The small-gap subtraction contributes shared statistical error between the classes; pair acceptance and tracking are shared between classes, while residual-level correlation is varied independently. A positive excess above the same three-standard-deviation threshold in either class falsifies the depleted sign. A finite zero constrains size but does not falsify an amplitude-free existence statement.

The comparator has no pp medium response: pp is the source's reference, and its MC without medium effects has no underlying-event variation with subleading location, "as expected" (J-093–J-094). FDT's additional source is the moving pattern's compressed-omnium response. There is no empty background and no distinct substance to remove in passing from PbPb to pp. The wake is the same geometrical adjustment to energy flow, read as a depletion opposite the jet. This single additional-source channel has existence code $\mathcal{E}_{\mathrm{FDT}} = +1$ and $\mathcal{E}_{\mathrm{ablated}} = 0$, with code threshold 1. It is not inferred from a coordinate relabelling. The test requires a pp integral precision beyond that established by the current qualitative flatness statement.

### 5.5 Numerical decision experiments

All numerical settings define exact synthetic distributions, not precision estimates of physical parameters. Inputs in this subsection are illustrative-assumed, except the pp integration interval and the systematic ranges drawn for pp, which the source reports; it names no system for those ranges, and its tracking range spans momentum, centrality and gap intervals. None is a measured hyperon property, a CMS count or an FDT amplitude. Repeated sampling concerns laboratory estimators, not a probabilistic ontology. The pseudorandom initialization is 20260930 and each scenario contains 20000 readings (U-24). The same nuisance calibration is redrawn for each reading and included in that reading's covariance. Each decision, including the polarization decision at both sample sizes, is tested at the comparator point and at the stated nonzero point.

For the polarization statistic, sample the normalized decay-angle law in Eq. (25) in 16 equal cosine bins (U-39). Let $\mathfrak{f}_{\mathrm{rej}}$ denote a rejection fraction. Let $\mathfrak{k}$ be the number of bins, $\mathfrak{c}_{\mathsf{b}}$ their centres and $\mathfrak{N}_{\mathsf{b}}$ their multinomial counts; here $\mathsf{b}$ indexes angular bins only. Let $\mathfrak{N}$ be the total reconstructed count. The operator Pr denotes classical sampling probability; hats in $\widehat{\mathcal{P}}^{\mathrm{tor}}$ and $\widehat{\mathfrak{a}}_{\mathrm{dec}}$ denote estimated polarization and calibrated decay parameter. The exact bin probabilities and the binned estimator, unbiased for a known decay parameter, are

$$\Pr(\mathsf{b}) = \frac{1 + \mathfrak{a}_{\mathrm{dec}} \mathcal{P}^{\mathrm{tor}} \mathfrak{c}_{\mathsf{b}}}{\mathfrak{k}}, \qquad \widehat{\mathcal{P}}^{\mathrm{tor}} = \frac{3 \sum_{\mathsf{b}} \mathfrak{N}_{\mathsf{b}} \mathfrak{c}_{\mathsf{b}}}{\mathfrak{N} \widehat{\mathfrak{a}}_{\mathrm{dec}} (1 - \mathfrak{k}^{-2})}. \tag{32}$$

This is exact sampling of binned decay angles. Its finite-bin factor follows by summing squared bin centres; the continuous estimator is recovered as the bin count increases. Calibration of the denominator is part of the repeated experiment, not held fixed. Synthetic decay parameters are $+0.75$ and $-0.75$ with calibration standard deviation 0.005; common vortical polarization is 0.02 (U-25–U-26). The nonzero point moves the species by $+0.03$ and $-0.03$ (U-12). Mean counts per species per bin are 2000 or 20000 (U-13), allocated in proportions $0.10, 0.15, 0.25, 0.30, 0.20$ to true-charge centres $-2, -1, 0, 1, 2$ (U-27–U-28). Charge resolution is 0.2 and is estimated from 100 random cones per bin per reading; latent bin means are calibrated by independent 400-cone samples (U-29, U-32). The intercept uses the calibrated means, so the measured regressor's attenuation is removed rather than ignored. The jet-charge polarization slope is 0.010 per unit charge; the shared detector offset is 0.010, with axis

mismatch standard deviation 0.002 (U-33). The three C-odd bounds are each 0.001 in the final species-difference statistic (U-34), assigned to spin-Hall, bulk current and fragmentation. The interior scenario draws each uniformly within its bound; the boundary scenario sets every term to its positive bound. Covariance uses one fully correlated shift, the sum of the three bounds, to cover co-oriented terms. These synthetic bounds stand for control-region calibrations; they are not source constraints. The charge fit subtracts the calibrated regressor variance from its weighted second moment; its intercept covariance includes the slope times the calibrated charge-mean error. The centre estimate uses the independent calibration sample, while the random-cone sample estimates that error variance on every reading.

For annulus counts, take Poisson samples with baseline mean 400000 per charge per true-charge bin in the inner annulus and an outer/inner mean ratio 1.1 (U-15, U-35). The effective bulk charge fraction combines a net-charge fraction 0.02 and a detector asymmetry 0.01; the bulk control count is 4000000 on average (U-36). The effective fraction is the sum of net-charge fraction and detector asymmetry divided by one plus their product. Its positive and negative counts are redrawn independently on every reading. The same charge resolution and calibration scheme applies. A charge slope of 200 contrast counts per unit charge, a surviving C-odd bound of 100 counts and leading leakage bound of 100 counts enter the zero-charge contrast (U-37). The nonzero scenario transfers fraction 0.002 of each inner yield, opposite ways for the two charges (U-14). The two bounds are sampled uniformly inside or set to their positive boundary; their covariance uses a fully correlated shift to cover co-oriented residuals. The intercept and product covariance include the bulk fluctuations and their signed-count covariance, the charge uncertainty and both nuisance terms.

For pp, subdivide the stated integration interval into 12 equal bins; each independent large-gap and shared small-gap statistical component has standard deviation 0.02 per bin (U-38). At each reading draw systematic amplitudes uniformly from the source ranges: residual 0.01–0.04, acceptance 0–0.01, tracking 0–0.025 (J-103, J-107, J-111), and multiply them by unit Gaussian pulls. Each pull shifts every bin coherently. Acceptance and tracking pulls are shared across classes; residual pulls have correlation 0, 0.5 or 1 (U-17). The stated covariance is recomputed from that reading's amplitudes. Nonzero integrated depths are 0.1, 0.2 and 0.5 per class (U-16). They assign no expected pp yield to the theory. The intentionally incorrect independent-bin calculation uses those same readings to expose the consequence of dropping correlations.

The table below gives repeated-sampling rejection fractions, with the nominal Gaussian benchmarks 0.0027 two-sided and 0.00135 one-sided (U-18–U-19). The entries are rounded to two significant figures; Monte Carlo uncertainty, not the printed decimal spacing, sets precision.

| Scenario | Rejection fraction |
|---|---|
| Polarization zero, interior bounds, smaller sample | 0.0027 |
| Polarization zero, all bounds, smaller sample | 0.0026 |
| Polarization nonzero, smaller sample | 0.039 |
| Polarization nonzero, larger sample | 0.80 |
| Polarization zero, interior bounds, larger sample | 0.0024 |
| Polarization zero, all bounds, larger sample | 0.0031 |
| Annulus zero, interior bounds | 0.0014 |
| Annulus zero, all bounds | 0.0027 |
| Annulus nonzero, interior bounds | 0.98 |
| PP zero, both negative, residual correlation zero | 0 |
| PP zero, both negative, residual correlation half | 0.00035 |
| PP zero, both negative, residual correlation one | 0.00055 |
| PP zero, either positive, maximum over correlations | 0.0027 |
| PP depth 0.1, half-correlated residual | 0.071 |
| PP depth 0.2, half-correlated residual | 0.55 |
| PP depth 0.5, half-correlated residual | 1 |
| PP zero, incorrect independent-bin single-class test | 0.12 |

These are conditional performance estimates of the specified experiment. An entry at either endpoint denotes an observed finite-ensemble frequency, not an exact population probability. Their sampling uncertainty away from those endpoints is the binomial standard deviation $\sqrt{\mathfrak{f}_{\mathrm{rej}}(1-\mathfrak{f}_{\mathrm{rej}})/20000}$. Real control-region bounds and covariance must replace the illustrative inputs before applying these power estimates to a sample. No ratio in the polarization statistic is treated as noise-free, and no missing reset component is held fixed at either hypothesis point.

### 5.6 Candidates without an observable map

DC-3, SA-07, is retired NO-MAP O-8 because the effective aperture supplies neither a constitutive sound speed nor a sound-cone opening-angle map. DC-4, SA-44, is retired NO-MAP O-5, with O-6, because the overlap evaluation and the charge/colour inputs needed for a driven force response are absent. DC-5, SA-02, is retired NO-MAP O-2 because no medium-state map constructs the A-3 odds or a closure crossing. DC-6, SA-33, is retired NO-MAP O-4 because no local-profile map returns the path-integrated centrality dependence.

## 6 Cross-domain reading

A black-hole jet is a moving pattern of the same omnium and carries a bow and wake. Its wake is the geometry resizing to its energy flow, a density disturbance. Read gravitationally, the density wave and gravity's ripple are one stroke (I-2), so the wake is a gravitational and density wave. The black hole is a hollow, double-sided density boundary: a dilute centre and saturated skin, with no singularity (0.6, 0.8). It carries its exterior atlas and an interior measured-support chart, with no transfer map between them. A black hole and an atomic nucleus are one construction read at different sizes.

For the same stress tensor and data, its wake's curvature is general relativity's by Eq. (6). The additional content is the carrier identity and the source response carried with the moving pattern. The CMS record contributes no astrophysical datum to this statement, and no

amplitude or length scale is transferred from its charged-hadron yields. Jet observations do not derive the open-endpoint condition of I-6. The density profile, the curvature solution and a detector's weighted observable remain separate objects at both sizes. Near a compact object the gravitational window of Eq. (17) is no longer small, and the same deviation equation reads the black-hole jet's wake in both windows.

# 7 System-level reading (exploratory)

This section re-describes the measured correlation in sample; it adds no prediction and assigns no new number to the source.

**X-1: displacement of the depletion.** The observable is the mean position of the negative near-side contrast $\Delta\mathcal{D}$ in 0–30% PbPb as the gap class increases (SA-37; J-147–J-150). Its defining condition is a count-weighted mean over the same negative signal window and track selection used by CMS, with long-range background subtracted. Its comparator is the corresponding mean in the smaller-gap class of the same data. The source reports, for the 0–30% PbPb-minus-pp difference at $1 < p_{\mathrm{T}}^{\mathrm{ch}} < 2\,\mathrm{GeV}$, that the mean position of the observed depletion shifts toward negative $\Delta\eta^{\mathrm{ch,j1}}$ as the gap selection increases from $(0.5, 1.0)$ to $(1.5, 2.0)$ (J-144, J-147–J-148), but prints no mean positions. Those endpoints describe a secant comparison, not a local derivative. In deviation form the reading is the jet-relative offset of each large-gap class, its mean depletion position plus its mean absolute dijet gap: rigid tracking keeps the offset constant, and its second difference across the three classes is the relative acceleration of wake and jet. The record prints neither mean, and no offset is assigned. In the Chladni reading the collision drives the density field and the counted hadrons mark its nodal architecture (P-2).

**Guards.** The weight is the final-state charged-hadron count per dijet pair and per radian in the near-side projection; no further interior weight is specified. No local density, overlap value, force or nodal solution is inferred. This is an in-sample identity-map reading, not corroboration of an invariant or a spectral target.

# 8 Convergence statement

On CMS's event, jet and track selections, corrections, near-side projection, background subtraction, normalization and observable dictionary, FDT returns the same measured distributions through Eq. (20). It accepts the central low-momentum significance as the reported lower bound, not an exactly reconstructed statistic. Model bands retain their simulation status and stated uncertainty scope. This limiting map is the identity of the data, corroborating no invariant. Containment belongs to I-3 at fixed stress tensor and data, and to I-4 L2 at leading weak-field, slow-motion order. For the same stress tensor and data the deviation of neighbouring patterns returns the tidal operator of general relativity, Eq. (17), and its area integral the I-3 deflection; the medium's patterns, read through Eq. (18), return the hydrodynamic computation on its own inputs.

The standard reading takes quarks and gluons as distinct particles traversing a separate medium and hadronizing to collimated sprays of particles (SA-01; J-007–J-009, J-017). FDT

reads hadrons as closed patterns on their own support, quarks as sub-braid strands and gluons as the one stroke standing inside a closed pattern, with no intrinsic mass or charge (0.1b, I-2, I-5).

The standard reading takes the QGP as a hot form of matter consisting of deconfined partons and colour interaction through gluon exchange (SA-08; J-018–J-020). FDT reads one state of the omnium, confinement as topological closure, colour as the SU(3) of the internal nodal surface of three-helix braids, and interaction as curvature plus geodesic motion (0.1, I-2, I-5).

The standard reading takes the modeled medium as a fluid evolving on a fixed spacetime background, into which the jet deposits energy (SA-49; J-026, J-151–J-152). FDT reads only omnium and its patterns, whose clocks and rulers adjust as the geometry itself, so the wake is spacetime moving with the energy flow and the hydrodynamic bands are computations on their own inputs (0.1, 0.1a, 0.1b). Read through Eq. (18), the medium's patterns free-fall on clocks and rulers rescaled together by the temperature, through the moving density structure the jet deposits, and the wake is visible only as the deviation of neighbouring patterns (0.1a, P-4).

FDT is a true extension of general relativity and quantum mechanics: it recovers these data on their own inputs and adds the circulation's magnetic projection or the moving source's mass shadow in the prospective tests. Neither compactness nor a branch label alone produces a divergence.

## 9 Status ledger

The table below states the mathematical and observational status once. The core labels retain their own meanings; a target is not a spectrum calculation and a provisional token is not an assigned value. In the mass targets, upright p, e and mu label proton, electron and muon rest masses. Electron and muon are the same three-helix braid; whether their difference is a higher box mode at fixed $n$ or a higher same-parity $n$ is the open fork, undetermined (I-5).

| Item | Status |
|---|---|
| O-1 | Owed: the finite-core cutoff $r_c$, the regular core chart, or the outward phase-transfer law. |
| O-2 | Owed: a geometrically constructed odds for the overlap-contrast application (A-3). |
| O-3 | Owed: a canonical odds for the confinement application (A-4). |
| O-4 | Owed: the local profile $\rho_{\mathrm{loc}}(r_f)$ from the omnium constitutive law with boundary, surface and core data, reducing to the measured nuclear profile. |
| O-5 | Owed: multipole and tidal remainders of the closed-form pair expression; the overlap complex. |
| O-6 | Owed: maps from a charge or colour seat to an I-1 input; derivation of the $A_2$ seat table from cone geometry; the gluon octet's dynamics. |
| O-8 | Owed: the constitutive derivation of $c_{\mathrm{eff}}$ and $\mathfrak{n}_{\mathrm{cone}}$ and its reconciliation with lensing. |
| O-9 | Owed: the confinement operator $D$, its domains, boundary data and spectrum; every mass ratio, including the two held-out targets. |
| O-14 | Owed: the mass-shadow response function and its closure; the release function of core 0.7, the mass-loss rate density and compression contrast of the unwinding bodies that feed the effective release term, named here and unused. It leaves the pp depletion size unspecified. |
| P-5 kernel | Conditional response form in Eq. (23); normalization and sign are unspecified, with nonzero projected overlap required for the two circulation tests. |
| A-3/A-4/A-5 | PROVISIONAL tokens and open range only; no value, threshold assignment, ordering or trajectory for this system. |
| Mass targets | Held-out $m_{\mathrm{p}}/m_{\mathrm{e}} = 1836.15267343$ and $(m_{\mathrm{mu}}/m_{\mathrm{e}})^2 = 4.2753 \times 10^4$ (U-05–U-06); neither is an output. |
| T-1 | Prospective, conditional by construction; opposite-species toroidal intercept; convergence value zero after the named controls. |
| T-2 | Prospective, conditional by construction; charge-odd annulus product; convergence value zero with the constant bulk-ratio condition and named bounds. |
| T-3 | Prospective, conditional by construction, single additional-source channel; both pp integrals negative; convergence value zero, no core amplitude. |
| Exploratory displacement | In-sample reported mean-position trend; no numerical mean, derivative or jet-relative offset assigned. |

# 10 Conclusions

The observed charged-hadron depletion and its significance bound survive unchanged on CMS's inputs and dictionary. FDT reads the wake as moving geometry of the one medium, with the same curvature as general relativity for the same stress tensor and data. The wake is visible only as the deviation of neighbouring patterns, the comparison that the source's gap and system subtractions make. The independent tests concern opposite-species toroidal polarization, charge-separated annulus transport and the pp depletion carried by the moving source. Their decisions include the shared calibration and systematic terms. None obtains a signal magnitude by fitting compactness to the existing depletion, and the amplitude-free null limits retain that distinction.

## References


[1] CMS Collaboration (A. Hayrapetyan et al.), *Observation of the Jet Diffusion Wake Using Dijets in Heavy-Ion Collisions*, Phys. Rev. Lett. **137**, 071902 (2026), DOI: 10.1103/g49y-8cjl.

[2] *Core FDT Invariants*, locked version 5.4, author lock 2026-09-13.

[3] Belle Collaboration, Phys. Rev. Lett. **122**, 042001 (2019), transverse hyperon polarization in fragmentation.